

SOFTWARE REQUIREMENTS SPECIFICATION

Class 4 Mathematics Digital Learning System

IEEE Std 830-1998 Compliant - 14 Section Format

Version 1.0

December 2025

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete description of requirements for the Golden Mathematics Class 4 Digital Learning System. It is intended for developers, project managers, testers, quality assurance personnel, educational content specialists, and stakeholders. This document follows IEEE Std 830-1998 guidelines and defines functional, non-functional requirements, system constraints, and interfaces for developing a comprehensive digital learning platform aligned with Ghana's NaCCA curriculum.

1.2 Scope

Product Name:

Golden Mathematics Class 4 Digital Learning System (GMLS)

The system will transform traditional mathematics education through interactive digital experiences covering all four NaCCA curriculum strands: Number Operations, Algebra (Patterns and Relations), Geometry and Measurement, and Data Handling. Benefits include improved student engagement, reduced teacher preparation time, enhanced parent involvement, scalable deployment, and data-driven decision-making.

1.3 Definitions, Acronyms, and Abbreviations

NaCCA: National Council for Curriculum and Assessment | GMLS: Golden Mathematics Learning System | SRS: Software Requirements Specification | IEEE: Institute of Electrical and Electronics Engineers | LMS: Learning Management System | API: Application Programming Interface | REST: Representational State Transfer | PWA: Progressive Web Application | SSO: Single Sign-On | WCAG: Web Content Accessibility Guidelines | AUCDT: AsanSka University College of Design and Technology

1.4 References

1. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
2. NaCCA Standard-Based Mathematics Curriculum for Basic Schools (2019)
3. Golden Mathematics for Basic Schools - Class 4 Textbook & Teacher's Guide
4. Ghana Data Protection Act, 2012 (Act 843)
5. WCAG 2.1 Level AA Standards

1.5 Overview

This document contains 14 sections following IEEE 830-1998: Introduction, Overall Description, Specific Requirements, External Interface Requirements, Functional Requirements Summary, Performance Requirements, Design Constraints, Software System Attributes, Logical Database Requirements, Other Requirements, Analysis Models, Supporting Information, Appendices, and Index.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The GMLS operates as a standalone educational platform with integration capabilities for school information systems, authentication services, cloud storage, and analytics platforms through REST APIs. The system provides distinct interfaces for students (child-friendly with gamification), teachers (comprehensive analytics), parents (progress monitoring), and administrators (system management).

2.2 Product Functions

- Curriculum Content Delivery: Interactive lessons for all four NaCCA strands with multimedia and cultural relevance
- Assessment and Evaluation: Formative and summative assessments with automated grading
- Progress Tracking: Individual and class-level analytics with predictive insights
- Classroom Management: Student rostering, assignment distribution, real-time monitoring
- Personalization: AI-powered adaptive learning with gamification elements

2.3 User Characteristics

Students (Ages 9-10): Primary users requiring intuitive, touch-friendly interfaces with minimal text. Teachers: Professional educators needing efficient classroom management with moderate technical skills. Parents: Diverse backgrounds interested in progress monitoring with basic smartphone proficiency. Administrators: School leaders with advanced technical skills for system configuration and reporting.

2.4 Constraints

- Regulatory: Ghana Data Protection Act compliance, NaCCA curriculum adherence, WCAG 2.1 Level AA
- Technical: Low-bandwidth support (2 Mbps minimum), offline functionality required
- Operational: 24/7 availability with 99.5% uptime, 1000+ concurrent users
- Budget: Open-source technology stack, mid-tier cloud infrastructure

2.5 Assumptions and Dependencies

Assumptions: Basic school internet connectivity, supervised device access, teacher digital literacy, parent smartphone access. Dependencies: NaCCA curriculum frameworks, SSO services, cloud infrastructure, payment gateways, email/SMS services, open-source library maintenance.

3. SPECIFIC REQUIREMENTS

3.1 Strand 1: Number Operations

Functional Requirements (10 total):

- FR-S1-001 to FR-S1-005: Counting, number recognition (0-10,000), place value with base-ten blocks, skip counting, comparison, number lines
- FR-S1-006 to FR-S1-010: Multi-digit addition/subtraction with regrouping, word problems, mental math strategies, fact fluency

3.2 Strand 2: Algebra (Patterns and Relations)

Functional Requirements (8 total):

- FR-S2-001 to FR-S2-004: Visual and numerical patterns, pattern creation tools, cultural patterns
- FR-S2-005 to FR-S2-008: Equality/inequality symbols, balance scale simulation, function tables, relational games

3.3 Strand 3: Geometry and Measurement

Functional Requirements (10 total):

- FR-S3-001 to FR-S3-005: 2D/3D shape identification, virtual manipulatives, properties, symmetry, transformations
- FR-S3-006 to FR-S3-010: Length, weight, capacity, time measurement tools, unit conversion, real-world problems

3.4 Strand 4: Data Handling

Functional Requirements (10 total):

- FR-S4-001 to FR-S4-004: Survey creation, data entry, frequency tables, collaborative projects
- FR-S4-005 to FR-S4-010: Bar graphs, pictographs, line plots, interactive exploration, interpretation, export

3.5 Cross-Cutting System Features

- Authentication (5 FRs): Registration, secure login, SSO, password recovery, session management
- Progress Tracking (5 FRs): Activity logging, performance metrics, dashboards, teacher analytics, reports
- Content Management (4 FRs): Curriculum alignment, versioning, multimedia integration, search
- Assessment System (5 FRs): Question bank, custom quizzes, automated grading, adaptive testing, feedback

4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

Student Interface: Large touch-friendly buttons (44x44px minimum), high-contrast colors, icon-based navigation, audio instructions. Teacher Interface: Comprehensive dashboard, drag-and-drop assignment creation, customizable views, multi-format exports. Parent Interface: Mobile-optimized progress view, notification center, messaging capability. Administrator Interface: System configuration panel, user management, aggregate reporting.

4.2 Hardware Interfaces

Standard input devices (touchscreen, keyboard, mouse), displays (1024x768 minimum to 4K), audio devices (headphones/speakers), cameras (QR scanning, document capture).

4.3 Software Interfaces

- School Information Systems: REST API with JSON for roster synchronization
- SSO Integration: SAML 2.0 and OAuth 2.0 protocol support
- Cloud Storage: Google Drive API v3, Microsoft OneDrive API v1.0
- Email/SMS: SMTP protocol, Twilio API for notifications

4.4 Communication Interfaces

HTTPS with TLS 1.3 for all communications, WebSocket for real-time features, RESTful API with JSON payloads for integrations.

5. FUNCTIONAL REQUIREMENTS SUMMARY

Total: 57 functional requirements across 8 categories. Strand 1: 10 FRs (6 high priority). Strand 2: 8 FRs (5 high priority). Strand 3: 10 FRs (7 high priority). Strand 4: 10 FRs (6 high priority). Authentication: 5 FRs (all critical/high). Progress Tracking: 5 FRs (4 high). Content Management: 4 FRs (3 high). Assessment: 5 FRs (4 high).

6. PERFORMANCE REQUIREMENTS

- Response Time: Page loads $\leq 3s$, interactions $\leq 100ms$, database queries $\leq 2s$ (95%), API $\leq 500ms$
- Capacity: 1000+ concurrent users, 10,000+ student records, 1 year online data, 100GB+ multimedia
- Reliability: 99.5% uptime (school hours), MTBF 720 hours, MTTR ≤ 15 minutes, data loss ≤ 2 minutes
- Availability: 24/7 with max 1 hour planned downtime/month, offline mode with auto-sync

7. DESIGN CONSTRAINTS

- Standards: WCAG 2.1 Level AA, ISO/IEC 25010:2011, Ghana Data Protection Act 2012
- Technology: React 18+ TypeScript, Spring Boot 3.2+, MySQL 8.0+/MariaDB 10.6+, Docker, Ubuntu 22.04 LTS

8. SOFTWARE SYSTEM ATTRIBUTES

- Security: Bcrypt encryption (10+ rounds), TLS 1.3, RBAC, SQL injection protection, XSS prevention
- Maintainability: 80%+ code coverage, modular architecture, comprehensive documentation, CI/CD
- Portability: Docker containerization, database abstraction, cloud-agnostic (AWS/Azure/GCP)
- Scalability: Horizontal scaling, read replicas, CDN for assets, auto-scaling

9. LOGICAL DATABASE REQUIREMENTS

Key Entities: Users, Students, Teachers, Classes, Content, Assessments, ActivityLogs, Achievements. Schema: Normalized relational model (3NF), foreign key constraints, strategic indexing. Integrity: Primary keys, referential integrity, check constraints, audit columns (created_at, updated_at, created_by, updated_by).

10. OTHER REQUIREMENTS

- Internationalization: English primary, Ghanaian languages (Twi, Ga, Ewe), UTF-8 encoding
- Legal: COPPA compliance, Data Protection Act, Terms of Service, Privacy Policy, parental consent
- Training: Teachers (4 hours), Administrators (2 days), Students (30 minutes), video tutorials, user guides

11. ANALYSIS MODELS

Primary Use Cases: Student completes lesson, Teacher creates assessment, Parent views progress, Administrator manages users, System tracks progress. Data Flow: Presentation → Business Logic → Database layers with encryption. State Models: Authentication, lesson progress, assessment, system operational states.

12. SUPPORTING INFORMATION

12.1 Implementation Phases

- Phase 1 (M1-3): Foundation - architecture, authentication, database
- Phase 2 (M4-6): Core Content - Strand 1, assessment, interfaces
- Phase 3 (M7-9): Extended - Strands 2-4, analytics, parent portal
- Phase 4 (M10-12): Enhancement - AI adaptation, gamification, offline mode

12.2 Testing & Deployment

Testing: Unit, integration, E2E, performance, security, accessibility, UAT. Deployment: Pilot (2-3 schools), feedback, staged rollout, ongoing monitoring.

13. APPENDICES

13.1 Glossary

Adaptive Learning:

AI-powered customization of content delivery based on student performance.

Cardinality:

Understanding that the last number counted represents total quantity in a set.

Formative Assessment:

Ongoing evaluation during learning to monitor progress and provide feedback.

Virtual Manipulatives:

Digital objects students manipulate on screen to explore mathematical concepts.

13.2 Revision History

Version 1.0 | December 2025 | AUCDT Development Team | Initial SRS release (IEEE 830-1998 compliant)

13.3 Approval Signatures

Project Manager

Date: _____

Lead Developer

Date: _____

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This index provides quick reference to key terms and sections. Numbers refer to section locations.

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